

Handout 3: Elasticity and Its Application

Mankiw Ch 5

Answer key

Problem 1. Your websites: how many fewer, and what happens to revenue

You build websites for local businesses. At \$200 a site you sell 12 a month. You are thinking about raising the price to \$250, and you expect to sell 8 a month at that price.

- a) Compute the price elasticity of demand by dividing by the starting values: first for the move from \$200 to \$250, then for the move from \$250 back to \$200. Why do the two answers differ?

\$200 → \$250: $\% \Delta P = 50/200 = 25\%$ and $\% \Delta Q = (8 - 12)/12 = -33.3\%$, so $E_d = 33.3/25 = \mathbf{1.33}$.

\$250 → \$200: $\% \Delta P = -50/250 = -20\%$ and $\% \Delta Q = (12 - 8)/8 = 50\%$, so $E_d = 50/20 = \mathbf{2.50}$.

Same two points, two answers. Each calculation divides by a different starting value, and with changes this large the base makes a big difference.

- b) Compute the elasticity by the midpoint method. Is your demand elastic or inelastic?

Divide by the averages, 10 sites and \$225: $\% \Delta Q = (8 - 12)/10 = -40\%$ and $\% \Delta P = (250 - 200)/225 = 22.2\%$, so $E_d = 40/22.2 = \mathbf{1.8}$. Greater than 1: demand is **elastic**. Quantity moves almost twice as much as price, in percent. The answer is the same in either direction, and it falls between the two answers in (a).

- c) Compute your total revenue at each price. Does the price increase raise revenue? Is that consistent with your answer to (b)?

At \$200: $200 \times 12 = \$2,400$. At \$250: $250 \times 8 = \$2,000$. Revenue **falls** by \$400. That is what elastic demand implies: price and revenue move in **opposite** directions, because the 40% fall in sites outweighs the 22% rise in price.

Revenue is not profit: four fewer sites also free up your time. Weighing that needs costs, which come later in the course.

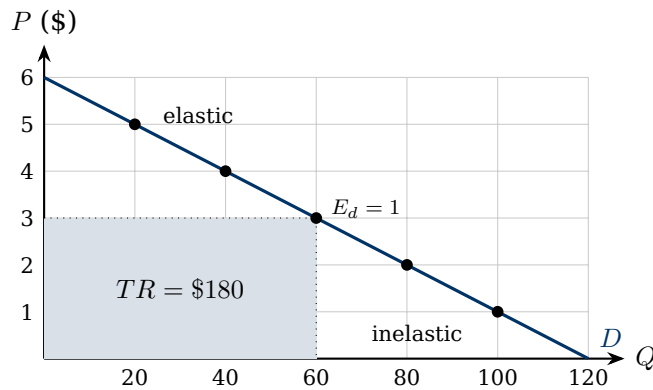
- d) Suppose instead that you would still sell 10 sites a month at \$250. Recompute the midpoint elasticity and the revenue. What changed?

$\% \Delta Q = (10 - 12)/11 = -18.2\%$ and $\% \Delta P = 22.2\%$, so $E_d = 18.2/22.2 = \mathbf{0.82}$: **inelastic**. Revenue is $250 \times 10 = \$2,500$, **up** \$100. Now price and revenue move **together**. The price change is identical; only the buyers' response changed. Demand this inelastic would mean your clients have few alternatives: few other designers nearby, or a site they need by next week.

Problem 2. Revenue along a straight-line demand curve

A taco truck near campus faces daily demand $Q_d = 120 - 20P$, where P is the price of a taco in dollars.

P (\$)	Q_d	$TR = P \times Q_d$ (\$)
1	100	100
2	80	160
3	60	180
4	40	160
5	20	100



- a) Fill in the table and draw the demand curve on the grid. At what price is total revenue largest?

At $P = \$3$: 60 tacos, $TR = \$180$. That is the **midpoint** of the line, which runs from \$6 on the price axis to 120 on the quantity axis.

- b) Compute the midpoint elasticity between \$1 and \$2, and between \$4 and \$5.

\$1 to \$2: $\% \Delta Q = -20/90 = -22.2\%$ and $\% \Delta P = 1/1.5 = 66.7\%$, so $E_d = \mathbf{0.33}$: inelastic.
 \$4 to \$5: $\% \Delta Q = -20/30 = -66.7\%$ and $\% \Delta P = 1/4.5 = 22.2\%$, so $E_d = \mathbf{3.00}$: elastic.
 For reference, the two middle steps: 0.71 between \$2 and \$3, and 1.40 between \$3 and \$4.

- c) The truck charges \$4 and wants more revenue. Should it raise or lower its price? What if it charges \$2?

At \$4 demand is elastic (1.40 on the step down to \$3), so price and revenue move in opposite directions: **lower** the price. Revenue rises from \$160 to \$180 at \$3. At \$2 demand is inelastic (0.71 on the step up to \$3), so price and revenue move together: **raise** the price. Revenue again rises from \$160 to \$180. From either side, revenue grows by moving toward the midpoint.

- d) Every step down this curve is the same: \$1 off the price, 20 more tacos. Why does the elasticity change along it?

The slope compares *units*: 20 tacos per dollar everywhere on the line. Elasticity compares *percentages*. At high prices, 20 tacos is a large percentage of the few tacos sold and \$1 is a small percentage of the price, so demand is elastic. At low prices it is the reverse. A straight line has one slope and every elasticity: elastic above the midpoint, inelastic below it.

Problem 3. Income, cross-price, and supply elasticities

For (a)–(e), compute the elasticity, keeping its sign, and classify the good or the pair of goods.

- a) Incomes rise 5 percent; purchases of store-brand cereal fall 2 percent.

$E_I = -2/5 = -\mathbf{0.4}$. Negative: an **inferior** good. As incomes rise, buyers switch to name brands.

- b) Incomes rise 5 percent; the number of international flights taken rises 12 percent.

$E_I = 12/5 = \mathbf{2.4}$. Positive and greater than 1: a **normal** good and a **luxury**.

- c) Incomes rise 5 percent; household electricity use rises 1 percent.

$E_I = 1/5 = \mathbf{0.2}$. Positive but small: a **normal** good and a **necessity**.

- d) The price of Uber rides rises 15 percent; the number of Lyft rides rises 9 percent.

$E_{xy} = 9/15 = \mathbf{0.6}$. Positive: **substitutes**.

- e) The price of game consoles falls 10 percent; sales of video games rise 5 percent.

$E_{xy} = 5/(-10) = -\mathbf{0.5}$. Negative: **complements**. Watch the signs: sales went up, but the price went down, so the ratio is negative.

- f) The price of fresh strawberries rises from \$3 to \$5 a pint, and within the week growers bring 1,100 pints to market instead of 900. Compute the price elasticity of supply by the midpoint method. Would you expect it to be larger or smaller if the price stayed at \$5 for three years? Why?

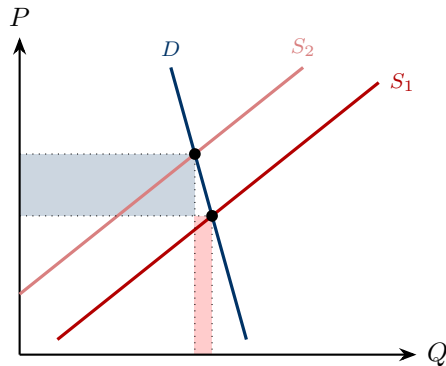
$\% \Delta Q = 200/1,000 = 20\%$ and $\% \Delta P = 2/4 = 50\%$, so $E_s = 20/50 = \mathbf{0.4}$: **inelastic**. Within a week, growers can only pick harder from the plants they already have. Over three years they can plant more acres and new growers can enter, so the same price brings out much more: the elasticity is **larger**. Supply is more elastic in the long run.

Problem 4. Two ways to fight drug use

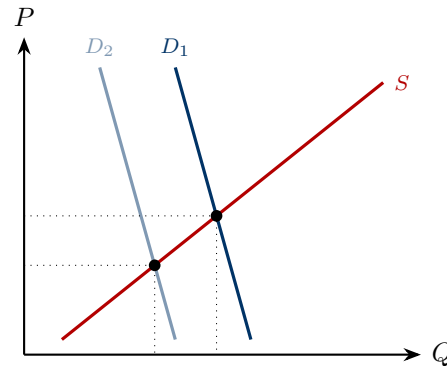
Two policies are proposed to reduce the use of illegal drugs:

- (1) Increase the number of federal agents devoted to drug interdiction.
- (2) Open more addiction treatment centers and spend more on education.

The demand for illegal drugs is highly inelastic. Use a diagram for each policy.



(1) spending rises



(2) spending falls

- a) Policy (1), interdiction. Which curve shifts, and which way? What happens to the price, the quantity, and total spending on drugs?

Seizing drugs raises the cost of bringing them to market: **supply shifts left** (S_1 to S_2), the price rises and the quantity falls. Demand is highly inelastic, so the price rises by a larger percentage than the quantity falls and spending $P \times Q$ **rises**: in panel (1) the blue area gained is larger than the red area lost. Dealers take in more.

- b) Policy (2), treatment and education. Same questions.

Fewer people want drugs at any price, so **demand shifts left** (D_1 to D_2). The price and the quantity both fall, so spending $P \times Q$ **falls** for certain, whatever the elasticity.

- c) Which policy reduces the total amount of money spent on drugs? Which is more likely to reduce drug-related crime?

Only **treatment and education** cuts spending; both policies cut the quantity used. Users who pay for drugs through crime need more money under interdiction, so drug-related crime can **rise**, at least in the short run. Under treatment and education it falls along with spending.

- d) Would your answer about interdiction change over a longer horizon?

It might. Over years demand is likely more elastic, because higher prices keep young people from starting. The more elastic demand is, the smaller the rise in spending; with elastic demand, interdiction would cut spending too.

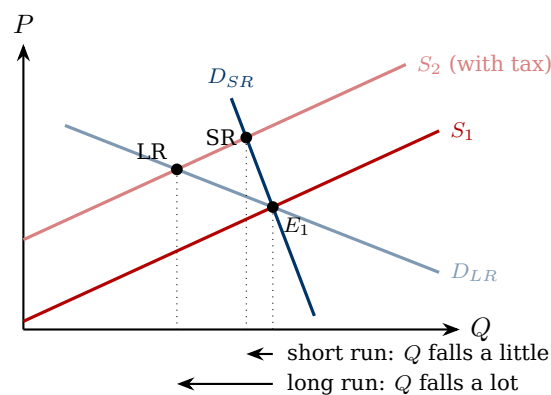
Problem 5. A tax to reduce smoking

Studies indicate that the price elasticity of demand for cigarettes is about 0.2. A pack currently costs \$5, and the government wants to reduce smoking.

- a) What price would reduce smoking by 10 percent?

$0.2 = 10/\% \Delta P$, so $\% \Delta P = 50\%$: the price must rise from \$5 to **\$7.50**. With inelastic demand, a small cut in quantity takes a large price increase. (The midpoint method applied to the price, $(P - 5)/[(P + 5)/2] = 0.5$, gives \$8.33: with a change this large the base matters, as in 1(a).)

- b) A tax raises the price of cigarettes. Is the policy more effective in the short run or in the long run? Show it on a supply-and-demand graph.



In the **long run**. The tax raises the cost of each pack: supply shifts up, S_1 to S_2 . In the short run smokers are addicted and slow to change, so demand is steep (D_{SR}) and the quantity falls a little. Over years smokers quit and fewer young people start, so demand is flatter (D_{LR}) and the quantity falls much more.

- c) Do teenagers have a higher price elasticity of demand for cigarettes than adults? Discuss using each statement.
- (i) It is legal for adults to drink alcohol, so many adults spend their money on alcohol rather than cigarettes.
 - (ii) Adults are more likely to be addicted to cigarettes.
 - (iii) Teenagers have less income than adults.

(i) Points the other way: adults have a legal substitute that teenagers do not, which on its own makes adults' demand *more* elastic.
 (ii) Addiction makes demand inelastic, and teenagers are less often addicted: **more** elastic.
 (iii) Cigarettes take a larger share of a smaller income: **more** elastic.
 On balance teenagers should be more price-sensitive, as most studies find: one reason cigarette taxes are defended as a way to reduce youth smoking.